Multi-Spectral Inpainting for Landsat-7 SLC-Off Image Reconstruction*

Michele Bilko
mbilko@bu.edu

Olivia Ma
oliviama@bu.edu

Lawrence Miao
ljmiao@bu.edu

Prem Rajendran
premraj@bu.edu

Abstract

The Landsat 7 satellite experienced a permanent failure of its Scan Line Corrector (SLC) in 2003, causing approximately 22% of each image to be lost in the form of stripe-shaped gaps. Although the remaining image content is still useful, these missing regions significantly reduce the scientific utility of the data for applications such as land-use monitoring, climate analysis, and environmental change detection. Restoring these corrupted regions accurately is therefore an important remote sensing problem. We approach this as a supervised image inpainting task, training on paired clean and corrupted Landsat patch pairs. We compare a Pix2Pix GAN baseline and an NSPI classical baseline against our modified LaMa model, which incorporates reference image guidance, with the goal of reconstructing missing regions in a way that preserves scientific and geographic accuracy rather than visual realism alone.

1 Related Work and Current Solutions

Previous work on Landsat-7 gap restoration includes both classical and deep-learning-based approaches. One line of work uses a clean reference image from the same location together with radiometric correction techniques such as histogram matching and eigendecomposition. Another line of work uses deep learning, including pix2pix-style GANs, to reconstruct stripe-corrupted satellite images. These methods provide useful starting points, but are limited by either strong assumptions about temporal consistency or by their tendency to prioritize visual realism over scientifically faithful reconstruction (Adıyaman et al., 2024; Case and Vitti, 2021).

A traditional non-deep-learning approach is the Neighborhood Similar Pixel Interpolator (NSPI), introduced by Chen et al. (2011) for filling gaps in Landsat ETM+ SLC-off imagery. NSPI uses a clean reference image of the same location from a different date and estimates each missing pixel by identifying spectrally similar neighboring pixels in the corrupted and reference images. This makes NSPI a useful classical baseline because it explicitly targets Landsat SLC-off gaps without requiring deep learning. However, because it relies on reference imagery and local spectral similarity assumptions, its performance can degrade when the reference image differs significantly from the corrupted image or when the missing regions contain complex spatial structure.

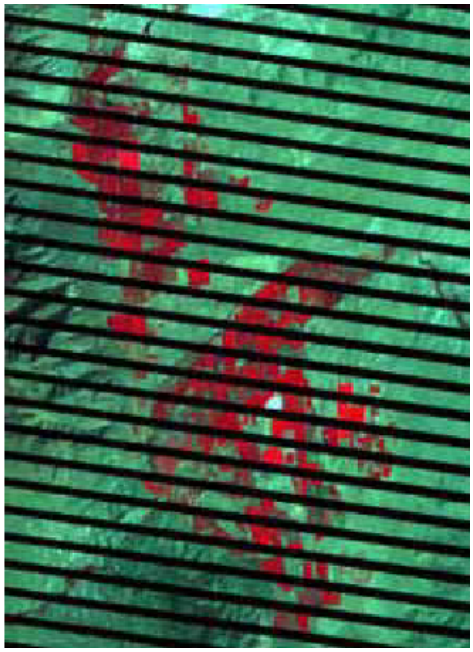


Figure 1: Corrupted image from Landsat-7 (Sanchez-Azofeifa, 2005).

The dataset used in the original Pix2Pix paper differs substantially from ours. Their dataset consists of clean and synthetically striped Landsat-8 patches stored as GeoTIFF files, using only the panchromatic band, with patch sizes such as 512×512 or 1024×1024, and 8-bit radiometric

*Code available at https://github.com/michele-bilko/landsat-slc-inpainting.

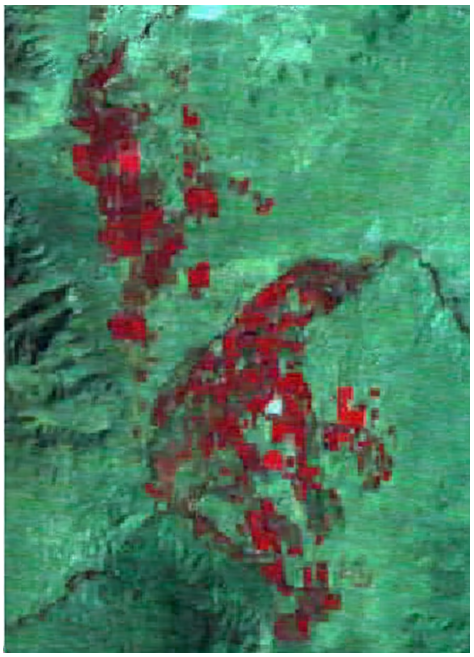


Figure 2: Adobe Firefly's generative fill on the gaps.

precision. In contrast, our dataset is designed for multi-spectral reconstruction: each sample is a 7-channel patch stored as a float32 NumPy array at 256×256 resolution.

2 Data

We address Landsat-style stripe-gap reconstruction as a supervised multi-spectral image inpainting task. Starting from raw Landsat GeoTIFF imagery, we extract the relevant spectral bands, convert each scene into a consistent 7-channel format, and crop the scenes into non-overlapping 256x256 patches stored as float32 NumPy arrays. We then simulate Landsat-7 SLC-off corruption using stripe patterns derived from real metadata, creating paired clean and corrupted samples for training. As a result, each example in our dataset consists of an aligned `clean.npy` and `corrupt.npy` pair, enabling the model to learn reconstruction of missing stripe regions directly from multi-channel satellite inputs.

3 Pre-Processing and Dataset Construction

We preprocess the data by first reading the raw Landsat GeoTIFF files with rasterio and extracting the spectral bands needed for our task. Since Landsat 7 uses 7 relevant channels while Landsat 8 and 9 contain more bands, we convert the imagery into a consistent 7-channel representation so that all training samples follow the same input format. After

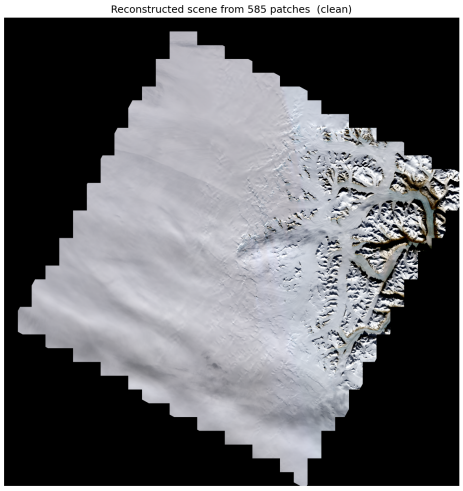


Figure 3: Reconstructed clean scene from extracted patches.

band selection, each scene is cropped into non-overlapping 256x256 patches. These patches are stored as float32 NumPy arrays in order to preserve more radiometric detail than an 8-bit image representation. All code, including dataset construction, evaluation, and baseline implementations, is available at https://github.com/michele-bilko/landsat-slc-inpainting.



Figure 4: Synthetic Landsat-7-style corruption applied to a reconstructed scene.

To create supervised training pairs, we simulate Landsat-7 SLC-off corruption using stripe geometry derived from real Landsat-7 metadata. This synthetic masking procedure zeros out approximately 22% of the pixels in a patch, matching the characteristic stripe-shaped missing regions observed in real SLC-off imagery. For each clean patch, we therefore generate a corresponding corrupted patch

and save the pair as `clean.npy` and `corrupt.npy`. This gives us aligned input-target pairs for training an inpainting model.

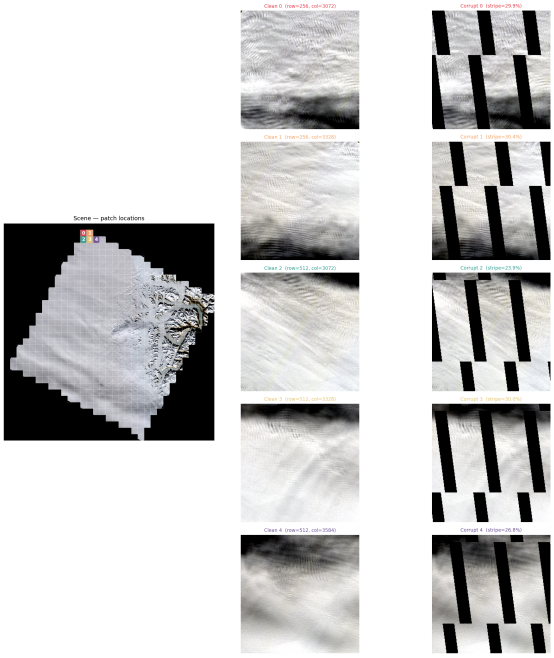


Figure 5: Patch extraction and sample generation pipeline.

If multiple Landsat sources are used, they are merged by harmonizing them into a common representation rather than by directly stitching images together pixel-by-pixel. In particular, Landsat 8 scenes are converted into the same 7-channel format used for our project, after which all scenes are patchified using the same 256x256 cropping procedure. Once they share the same channel structure, spatial size, numeric precision, and corruption process, the resulting patches can be combined into a single unified training set. We do not merge the original paper's dataset into our main training pipeline directly. Instead, we treat it as a comparison resource because it uses a different representation: single-channel panchromatic GeoTIFF patches rather than multi-channel float32 NumPy patches.

4 Experimental Setup

We formulate our task as a supervised multi-spectral image inpainting problem in which the input is a 7-channel satellite patch corrupted by Landsat-7-style SLC-off stripe gaps and the target is the corresponding clean 7-channel patch. The stripe mask has a highly structured geometry rather than a random missing region, so the model must

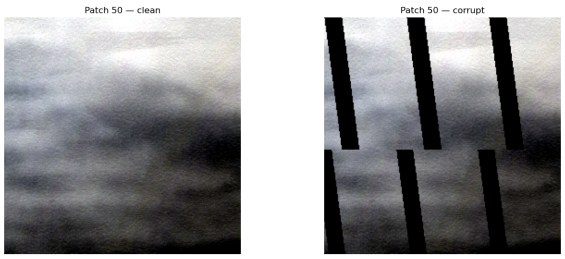


Figure 6: Example clean and corrupted patch pair.

learn to reconstruct spatially coherent terrain features under realistic Landsat corruption patterns.

To support efficient training, we store the preprocessed data as paired `.npy` files rather than reading raw GeoTIFF scenes during each training iteration. Each sample consists of a corrupted 7-channel patch and its corresponding clean target patch, saved in a format that can be loaded directly through `torch.utils.data.Dataset`. This design reduces preprocessing overhead at training time, simplifies batching in PyTorch, and ensures that all samples share a consistent shape, channel structure, and numeric precision. Compared with the original paper's dataset, which uses single-channel panchromatic GeoTIFF patches, our dataset is explicitly designed for multi-spectral learning with float32 256x256 patches, making it better suited for models that can exploit cross-channel information during reconstruction.

5 Baselines

We evaluate our implementation against a Pix2Pix GAN baseline and compare two variants of our modified LaMa model to isolate the contribution of reference image guidance. All models are trained and evaluated on the same dataset split at a resolution of 256×256 pixels.

The corrupted image is provided as the conditioning input and the model is trained to generate the inpainted image. This serves as our primary baseline, as it represents the most recent deep learning approach applied to the Landsat-7 inpainting problem (Adıyaman et al., 2024).

We additionally evaluate NSPI (Neighborhood Similar Pixel Interpolator) as a non-deep-learning baseline. For each corrupted patch, NSPI is provided with a clean reference image of the same location from a different timestamp. It fills missing pixels by identifying spectrally similar neighbors in the reference image and blending their values into the gap. This baseline represents the class of

classical reference-based methods and provides a point of comparison that does not rely on learned representations.

6 LaMa Architecture

Our model builds on LaMa (Large Mask Inpainting), introduced by Suvorov et al. (2022), which achieved state-of-the-art results on inpainting tasks involving large missing regions. LaMa is a convolutional encoder–decoder network trained to reconstruct masked regions from a corrupted input and a mask. Its key architectural innovation is the use of Fast Fourier Convolution (FFC) blocks in place of standard residual blocks in the network's bottleneck, which gives the model a global receptive field at every layer.

The Landsat-7 SLC-off mask pattern fits this setting closely. The stripe gaps are distributed across the entire scene rather than localized to one region, and reconstructing spatially coherent terrain features such as rivers, roads, and field boundaries requires global information about scene structure. LaMa's global receptive field allows it to propagate this structural information across the full image in a single forward pass, making it a natural starting point for our task. We extend it in two ways, described in subsequent sections: a temporally offset reference image as an additional input pathway.

We evaluate two variants of the LaMa model, both trained with our updated loss function:

- **LaMa (ours, no reference)**: The base LaMa architecture modified with our loss function, taking only the corrupted image and stripe mask as input.
- **LaMa (ours, reference)**: The full proposed model, which additionally takes an unmasked image of the same location at a different timestamp as a third input pathway, guiding reconstruction with known physical features.

7 Results

All models are evaluated on the same 256×256 patches using four metrics computed exclusively on the masked (stripe) region: PSNR, SSIM, CC, and RMSE. Restricting our evaluation to only the corrupted pixels directly measures reconstruction quality where data was missing, rather than diluting the scores with regions the model never needed to touch. All patches are normalized to [0, 1] per channel before metric computation.

Model	PSNR ↑	SSIM ↑	CC ↑	RMSE ↓
NSPI	20.01	0.429	0.646	0.1035
Pix2Pix	20.53	0.485	0.715	0.0980
LaMa (no ref)	25.97	0.903	0.950	0.0505
LaMa (with ref)	**33.93**	**0.972**	**0.991**	**0.0205**

Table 1: Quantitative comparison on the Landsat-8 test set. All metrics computed on masked (stripe) pixels only, with patches normalized to [0,1].

Table 1 reports results across all four methods. LaMa with reference image guidance achieves the strongest performance across all four metrics, with a PSNR of 33.93 dB, SSIM of 0.972, CC of 0.991, and RMSE of 0.0205. The no-reference LaMa variant achieves a PSNR of 25.97 dB, SSIM of 0.903, CC of 0.950, and RMSE of 0.0505. Pix2Pix and NSPI score substantially lower, with Pix2Pix at 20.53 dB PSNR and 0.485 SSIM, and NSPI at 20.01 dB PSNR and 0.429 SSIM.

The gap between LaMa with and without the reference image, roughly 8 dB in PSNR and 0.07 in SSIM, demonstrates that our temporal reference guidance really does improve reconstruction quality. The particularly low SSIM scores for Pix2Pix (0.485) and NSPI (0.429) indicate that while both methods approximate rough brightness values in the stripe region, neither reliably reconstructs correct spatial structure and texture. We also note that the no-reference LaMa variant already outperforms both Pix2Pix and NSPI on every metric. This suggests that the deep learning approach captures terrain structure more effectively than traditional interpolation-based methods, even without access to a clean reference image.

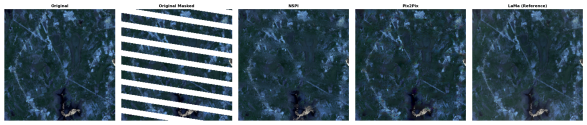


Figure 7: Visual comparison between implementations.

8 Interpretability Study

We conduct an interpretability study on our model by evaluating whether the model is properly exploiting the reference image. We conduct a gradient-based attribution analysis that visualizes which regions of the reference image influence the inpainted output.

8.1 Method

We use vanilla input gradients (Simonyan et al., 2013) to attribute the reconstruction loss back to the reference image. We define the target scalar as the mean L_1 reconstruction loss inside the stripe-shaped hole region, averaged over our 7 spectral channels, and compute its gradient with respect to the reference image input via a single backward pass. We collapse the channel dimension of the resulting tensor by taking the per-pixel L_2 norm to produce a 2D attribution map, which quantifies how sensitive the in-hole reconstruction is to small perturbations at each spatial location of the reference.

8.2 Visualization Results

Figure 8 shows the attribution map on the reference image for a representative test sample. It appears that the gradient magnitude is concentrated at the locations corresponding to the stripe gaps in the masked input, with comparatively low magnitude elsewhere. This means that the model uses the reference in a spatially localized manner. When reconstructing a hole pixel, it primarily draws information from the same coordinate in the reference rather than aggregating globally. This result supports our design rationale: a model that ignored the reference, or used it only as a global style condition, would have produced an attribution map uncorrelated with the mask structure.

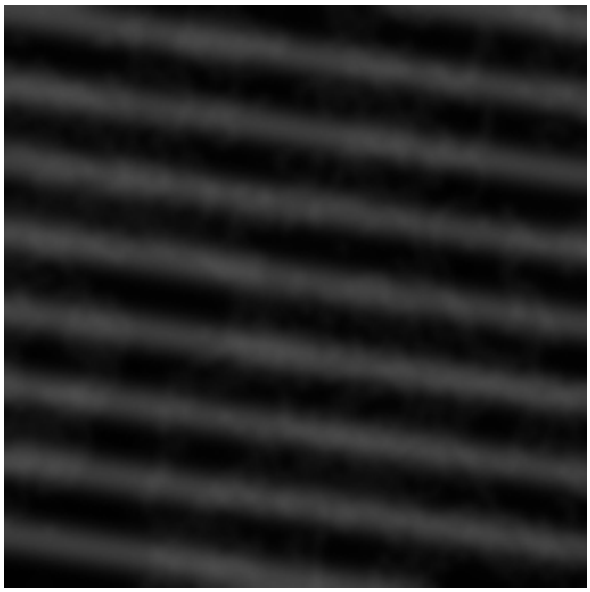


Figure 8: Vanilla input gradient attribution map on the reference image. Gradient magnitude is highest around the locations corresponding to the masked region.

9 Conclusion

9.1 Summary

We addressed the problem of inpainting striped gaps in Landsat-7 SLC-off imagery, framing reconstruction as a multi-spectral image inpainting task with the goal of preserving scientific and geographic accuracy rather than visual realism or aesthetics alone. We constructed a dataset of paired clean and corrupted 7-channel Landsat-8 patches at 256×256 resolution, with synthetic SLC-off corruption derived from real Landsat-7 stripe geometry, and additionally curated temporal reference patches for methods that exploit a clean image of the same location at a different date.

We compared three reconstruction methods on a shared evaluation set: NSPI as a non-deep-learning reference-based baseline, the pix2pix GAN as our deep learning baseline, LaMa without reference image guidance, and our proposed LaMa variant with reference image guidance. Our reference-guided LaMa achieved 33.93 dB PSNR and 0.972 SSIM, substantially outperforming both the no-reference variant (25.97 dB, 0.903) and NSPI (20.01 dB, 0.429). The roughly 8 dB gap between the two LaMa variants demonstrates that temporal reference guidance is the dominant contributor to reconstruction quality, while NSPI's lower scores confirm that classical reference-based interpolation cannot recover spatial structure as effectively as a learned model. These results validate the central design choices of our approach.

We applied NSPI to our evaluation patches using a clean Landsat-8 image of the same location acquired at a different date as the reference. The algorithm fills each gap pixel by combining a spatial prediction (a weighted average of nearby valid target pixels) with a temporal prediction (the reference value adjusted by the locally-estimated radiometric change between dates), merged by an uncertainty-weighted average that favors the more confident component in each region. Our implementation is available in the project repository.

9.2 Future Work

Real Landsat-7 evaluation: Our quantitative evaluation has been conducted exclusively on Landsat-8 imagery with synthetic SLC-off corruption, since this provides the paired ground truth required for metric computation. The next step would be to apply our method to actual Landsat-7 SLC-off scenes and compare reconstructed outputs against Landsat-

8 or Landsat-9 imagery of the same locations as a proxy for ground truth. This evaluation would test whether the model generalizes from its synthetic training distribution to the true sensor noise, atmospheric conditions, and stripe geometry of real Landsat-7 data, providing a stronger measure of practical utility than purely synthetic evaluation can.

References

Hilal Adıyaman, Yunus Emre Varul, Tolga Bakırman, and Bülent Bayram. 2024. Stripe error correction for landsat-7 using deep learning. *PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science*, 93:51–63.

Nicola Case and Alfonso Vitti. 2021. Reconstruction of multi-temporal satellite imagery by coupling variational segmentation and radiometric analysis. *ISPRS International Journal of Geo-Information*, 10:17–17.

Jin Chen, Xiaolin Zhu, James E. Vogelmann, Feng Gao, and Suming Jin. 2011. A simple and effective method for filling gaps in landsat etm+ slc-off images. *Remote Sensing of Environment*, 115(4):1053–1064.

Gerardo Arturo Sanchez-Azofeifa. 2005. Effect of scan line corrector fault on landsat imagery.

Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. 2013. Deep inside convolutional networks: Visualising image classification models and saliency maps. *arXiv preprint arXiv:1312.6034*.

Roman Suvorov, Elizaveta Logacheva, Anton Mashikhin, Anastasia Remizova, Arsenii Ashukha, Aleksei Silvestrov, Naejin Kong, Harshith Goka, Kiwoong Park, and Victor Lempitsky. 2022. Resolution-robust large mask inpainting with fourier convolutions. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 2149–2159.